

The if-then and if-then-else Statements

The if-then Statement

The `if-then` statement is the most basic of all the control flow statements. It tells your program to execute a certain section of code *only if* a particular test evaluates to `true`. For example, the `Bicycle` class could allow the brakes to decrease the bicycle's speed *only if* the bicycle is already in motion. One possible implementation of the `applyBrakes` method could be as follows:

```
void applyBrakes() {
    // the "if" clause: bicycle must be moving
    if (isMoving){
        // the "then" clause: decrease current speed
        currentSpeed--;
    }
}
```

If this test evaluates to `false` (meaning that the bicycle is not in motion), control jumps to the end of the `if-then` statement.

In addition, the opening and closing braces are optional, provided that the "then" clause contains only one statement:

```
void applyBrakes() {
    // same as above, but without braces
    if (isMoving)
        currentSpeed--;
}
```

Deciding when to omit the braces is a matter of personal taste. Omitting them can make the code more brittle. If a second statement is later added to the "then" clause, a common mistake would be forgetting to add the newly required braces. The compiler cannot catch this sort of error; you'll just get the wrong results.

The if-then-else Statement

The `if-then-else` statement provides a secondary path of execution when an "if" clause evaluates to `false`. You could use an `if-then-else` statement in the `applyBrakes` method to take some action if the brakes are applied when the bicycle is not in motion. In this case, the action is to simply print an error message stating that the bicycle has already stopped.

```
void applyBrakes() {
    if (isMoving) {
        currentSpeed--;
    } else {
        System.err.println("The bicycle has already stopped!");
    }
}
```

The following program, `IfElseDemo`, assigns a grade based on the value of a test score: an A for a score of 90% or above, a B for a score of 80% or above, and so on.

```
class IfElseDemo {
    public static void main(String[] args) {

        int testscore = 76;
        char grade;

        if (testscore >= 90) {
            grade = 'A';
        } else if (testscore >= 80) {
            grade = 'B';
        } else if (testscore >= 70) {
            grade = 'C';
        } else if (testscore >= 60) {
            grade = 'D';
        } else {
            grade = 'F';
        }
        System.out.println("Grade = " + grade);
    }
}
```

The output from the program is:

```
Grade = C
```

You may have noticed that the value of `testscore` can satisfy more than one expression in the compound statement: `76 >= 70` and `76 >= 60`. However, once a condition is satisfied, the appropriate statements are executed (`grade = 'C';`) and the remaining conditions are not evaluated.